

Computational Analysis of Gender Norms on Nairaland

Summary of Methods and Findings

Project: MAPGEN – Measuring and Analyzing Portrayals of Gender Norms
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Executive Summary

This report presents the results of a computational analysis of gender-related social norms expressed on Nairaland, one of Africa’s largest online forums, with over 3 million registered users in Nigeria. Using a novel pipeline combining large language models (LLMs) with sentence-embedding clustering, the analysis extracted and scored over 31,000 unique beliefs from 146,000 forum posts, organizing them into 740 thematic norm clusters. Each belief was rated for its restrictiveness (limitation of decision-making freedom) and burden (severity of sanctions for non-compliance), separately for men and women.

The central finding is a pervasive gendered asymmetry: all 38 coarse norm themes identified in the data are more restrictive for women than for men. Across the full set of 740 fine-grained clusters, 58% primarily constrain women, while only 13% primarily constrain men. Norms affecting women are also substantially more prescriptive: 84.6% are classified as injunctive (dictating what women ought to do), compared to 72.6% for men. The domains with the largest gender gaps involve bodily autonomy and appearance (hijab, makeup, modest dress), domestic roles (cooking, household responsibilities), and sexuality and reproduction (virginity, breastfeeding, premarital sex). Norms that disproportionately constrain men cluster around expectations of masculinity, fatherhood, and sexual performance.

This work also constitutes a methodological contribution, demonstrating that LLMs can be used at scale to surface, annotate, and quantify social norms from unstructured online text, a task that has traditionally required expensive, small-sample survey work or qualitative research. The accompanying data appendix provides a public resource for researchers and practitioners working on gender norms in Nigeria and West Africa.

1. Background

Social norms, the informal rules governing what is considered acceptable behavior in a community, are among the most powerful but hardest-to-measure determinants of gender inequality. Traditional approaches to measuring norms rely on surveys, focus groups, and ethnographic methods, which are costly, geographically limited, and subject to social desirability bias. As part of the MAPGEN (Measuring and Analyzing Portrayals of Gender Norms) initiative,

we explored whether large language models could serve as a complementary tool for norms measurement at scale.

The analysis focused on Nairaland (nairaland.com), founded in 2005 by Seun Osewa. With over 3 million registered members and approximately 400,000 daily visitors, Nairaland is one of Africa’s largest online forums. Its discussions span politics, religion, culture, entertainment, and family life, and its simple, text-based interface is optimized for low-bandwidth access, making it broadly accessible across Nigeria. The platform’s grassroots character and candid discourse make it a uniquely rich source of organically expressed social attitudes, beliefs, and norms.

2. Data and Methodology

The analysis proceeded through a multi-stage pipeline:

Data Collection. Approximately 4 million forum posts spanning seven years were collected from Nairaland, covering a broad range of discussion topics.

Filtering. A two-step filtering process was applied. First, a dictionary of sanction-related key phrases (derived from qualitative social norms research in Nigeria) reduced the corpus to approximately 300,000 posts. Second, GPT-4o-mini was used to identify posts referencing social situations containing explicit or implicit social norms, yielding a working dataset of 146,000 posts.

Belief Annotation. GPT-4o was used to annotate each of the 146,000 posts along several dimensions: the underlying belief or norm expressed, the behavior referenced, the social actor involved, and the associated social emotion. A validation sample of 50 posts was independently reviewed by three Nigerian annotators, achieving 96–100% agreement across attributes.

Restrictiveness and Burden Scoring. Each annotated belief was evaluated along two dimensions, separately for men and women.

- **Restrictiveness** measured the degree to which the norm limits decision-making freedom (1–5 scale).
- **Burden** measured the severity of punitive consequences for non-compliance (1–5 scale).

The model also classified each norm as injunctive (prescribing what people ought to do) or descriptive (reflecting what people actually do).

Clustering. Sentence-BERT embeddings were computed for each extracted belief, and agglomerative clustering (distance threshold = 5) produced 740 fine-grained norm topic clusters. These were further consolidated into approximately 40 coarse thematic groups for interpretive analysis.

Table 1. Pipeline Summary Statistics

Metric	Value
Total Nairaland posts collected	~4,000,000
Posts after dictionary filtering	~300,000
Posts after LLM norm detection	~146,000
Unique beliefs annotated	31,456
Fine-grained norm clusters	740

Coarse thematic groups	38
Annotator agreement (validation, n=50)	96–100%

3. Key Findings

Gendered Asymmetry

Across all 740 norm clusters, a clear gendered asymmetry emerged: 58% of clusters primarily constrained women, 13% primarily constrained men, and 30% were gender-neutral. All 38 coarse norm themes are more restrictive for women than for men. No coarse theme was identified in which men face more restrictive norms on average than women. Across the 31,456 annotated beliefs, women's average restrictiveness score was 3.12 (vs. 2.68 for men) and average burden score was 2.42 (vs. 2.07 for men). Norms affecting women were far more prescriptive: 84.6% were classified as injunctive, compared to 72.6% for men (Table 2).

Table 2. Average Restrictiveness and Burden Scores by Gender

Measure	Men	Women	Difference (W–M)
Avg. Restrictiveness Score	2.68	3.12	+0.44
Avg. Burden Score	2.07	2.42	+0.35
% Injunctive Norms	72.6%	84.6%	+12.0 pp
% Descriptive Norms	27.4%	15.4%	–12.0 pp

The distributions of restrictiveness and burden scores (Figure 1) show that women's scores are shifted toward higher values on both measures. The rightward skew in the gender difference distributions (Figure 2) confirms that the large majority of beliefs impose greater constraints on women.

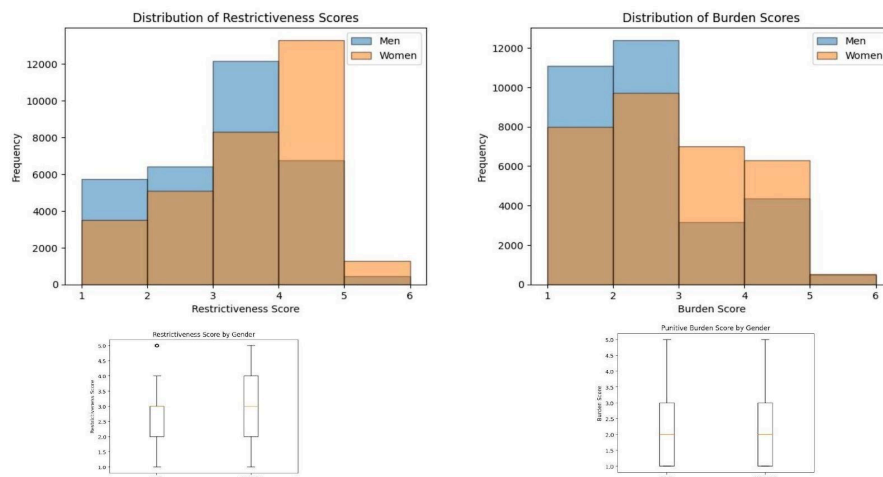


Figure 1. Distributions of restrictiveness and burden scores by gender, with box plots.

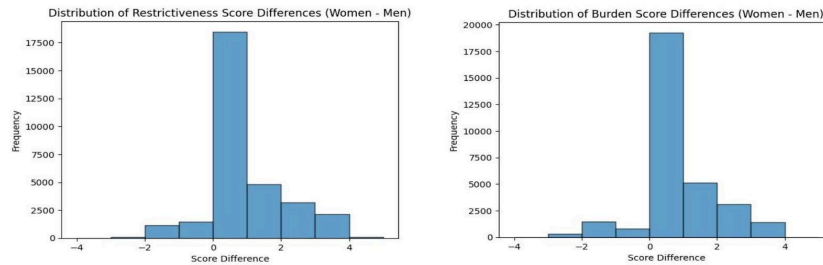


Figure 2. Distribution of score differences (Women – Men) for restrictiveness and burden.

Where Gender Gaps Are Largest

To move beyond aggregate statistics, we examined which specific normative domains exhibit the largest gender differentials. The norm clusters with the highest differential restrictiveness for women included topics related to hijab and modest dressing, perceptions of makeup and natural beauty, gender roles in religious contexts, and breastfeeding practices. In the most extreme cases, norms were up to 2.4 times more restrictive for women than for men. Conversely, the norm clusters with the highest differential impact on men related to expectations of masculinity, the role of fathers in family dynamics, and sexual performance expectations. Overall, results reveal five broad domains of gendered constraint:

Bodily Autonomy and Appearance. Norms governing women’s physical presentation exhibit the largest gender gaps in the entire dataset. Beliefs about makeup and natural beauty show a restrictiveness differential of +2.42 (women vs. men), while hijab-related norms show a differential of +2.32 with the highest burden gap of +2.14. Other clusters in this domain include modesty in dress (+0.86), women’s attire and clothing norms (+1.32), and wedding attire expectations (+1.45). These norms prescribe how women’s bodies should be presented in public and carry real social penalties for non-compliance — ranging from gossip and ridicule to religious censure.

Domestic Roles and Marriage. The expectation that women bear primary responsibility for cooking, cleaning, and childcare is one of the most frequently discussed norms on Nairaland. The “Responsibilities of Women in Family” cluster ($n=1,017$ beliefs) carries a restrictiveness gap of +1.10. “Gender Roles in Cooking and Domestic Responsibilities” (+1.42) is even more sharply gendered. Representative beliefs include “women are expected to cook for their husbands” and “inability to cook is frowned upon and can lead to social consequences.” The surname traditions cluster (+1.34) reveals that women are expected to take their husband’s name, with not doing so characterized as “disrespectful or nonsensical.”

Religion and Gender. Religious interpretation intersects powerfully with gender norms. The “Islam and Gender” coarse theme is the 7th most frequent ($n=1,318$) and carries a substantial restrictiveness gap of +0.85. Finer-grained clusters include “Gender Roles in Religious Contexts” (+1.77) and “Menstruation and Religious Practices” (+1.63), where beliefs such as “menstruating women are considered spiritually impure, leading to restrictions” impose gender-specific constraints rooted in religious authority.

Sexuality and Reproduction. Norms around virginity, premarital sex, and reproductive behavior are both frequent and sharply gendered. The “Value of Virginity in Relationships” cluster (+1.53 restrictiveness, +1.71 burden) contains beliefs such as “virginity is valued and expected for women” alongside counter-norms like “virginity is a social construct used to control behavior.” The “Premarital Sex” coarse theme (n=1,403) is one of the most common discussion topics and carries a +0.51 restrictiveness gap. Breastfeeding norms (+1.64) and pregnancy care norms (+1.27) further constrain women’s bodily autonomy in the reproductive domain.

Norms That Constrain Men. While the overall asymmetry strongly favors constraints on women, a small set of norms disproportionately burden men. “Expectations of Masculinity” carries a restrictiveness gap of –1.06 (favoring constraint on men), with beliefs such as “men are expected to be decision makers and responsible for their actions.” “The Role of Fathers in Family Dynamics” (–0.96) and “Sexual Performance and Health Management” (–0.77) are the other clusters where men face distinctly greater normative pressure. These norms center on men’s expected roles as providers, protectors, and sexually capable partners.

Table 3. Norm Clusters with Largest Gender Differentials

Cluster Theme	Restrict. (W–M)	Burden (W–M)	N	Direction
Perceptions of Makeup and Natural Beauty	+2.42	+1.13	125	Women
The Significance of Hijab in Islam	+2.32	+2.14	311	Women
Gender Roles in Religious Contexts	+1.77	+1.56	230	Women
Breastfeeding Practices and Societal Perspectives	+1.64	+0.86	58	Women
Menstruation and Religious Practices	+1.63	+0.99	125	Women
The Value of Virginity in Relationships	+1.53	+1.71	217	Women
Wedding Attire and Cultural Norms	+1.45	+0.68	110	Women
Gender Roles in Cooking and Domestic Responsibilities	+1.42	+0.98	171	Women
Women’s Behavior and Societal Expectations	+1.35	+1.13	175	Women
Surname Traditions and Identity in Marriage	+1.34	+0.69	96	Women
Expectations of Masculinity	–1.06	–1.17	304	Men
The Role of Fathers in Family Dynamics	–0.96	–0.93	168	Men
Sexual Performance and Health Management	–0.77	–0.51	78	Men

Note: Positive values indicate higher restrictiveness/burden for women; negative values indicate higher impact on men.

Most Frequently Discussed Norms

The most commonly discussed norms on Nairaland are not necessarily those with the largest gender differentials. Table 4 shows the ten most frequent coarse themes. Relationship dynamics, sexual ethics, and family matters dominate the discourse. Notably, every one of these high-frequency themes is more restrictive for women, though the magnitude of the gap varies — from +0.13 (rape and sexual violence, where norms constrain both genders almost equally) to +0.97 (physical attractiveness, where women bear a substantially higher burden of appearance-related expectations).

Table 4. Ten Most Frequently Discussed Coarse Norm Themes

Coarse Theme	N	R (W)	R (M)	R (W–M)	B (W–M)
Romantic Relationships	1,812	3.11	2.65	+0.46	+0.33
Serious Relationships & Marriage	1,640	2.86	2.64	+0.21	+0.17
Sexual Ethics and Norms	1,639	3.22	2.90	+0.31	+0.30
Family / Domestic Issues	1,475	3.15	2.70	+0.45	+0.31
Marriage as a Life Goal	1,426	3.09	2.68	+0.41	+0.36
Premarital Sex	1,403	3.23	2.72	+0.51	+0.56
Islam and Gender	1,318	3.78	2.93	+0.85	+0.79
Rape / Sexual Violence	1,220	3.05	2.92	+0.13	+0.11
Responsibilities in Marriage	1,150	2.95	2.74	+0.21	+0.18
Physical Attractiveness	1,070	3.14	2.18	+0.97	+0.64

R = Restrictiveness; B = Burden; W = Women; M = Men. All scores on 1–5 scale.

Priority Areas

Beliefs were also classified into priority policy domains. Among the 31,456 unique beliefs analyzed, 1,043 were associated with gender-based violence, 411 with child marriage, 355 with sexual and reproductive health, and 303 with women's economic empowerment.

4. Methodological Contributions

This work demonstrates a scalable approach to extracting and quantifying social norms from large-scale, unstructured, user-generated text. The pipeline combines dictionary-based filtering, LLM-based norm detection and annotation, sentence-embedding clustering, and gendered scoring along theoretically grounded dimensions (restrictiveness and burden). The methodology was developed in close consultation with social norms researchers at the World Bank, drawing on established survey frameworks such as the “out of 10” perceived sanctions format used in field research, as well as social norms taxonomies and measurement guides from the broader literature.

5. Public Data Resource

The accompanying appendix (Table A1) provides a summary of all 38 coarse norm themes identified in the Nairaland dataset, including frequency counts, average restrictiveness and burden scores by gender, and illustrative GPT-generated theme descriptions. This resource is intended to support researchers and practitioners working on gender norms in Nigeria and West Africa, offering a computationally derived landscape of the norms that shape online discourse — and, by extension, the social attitudes of a significant segment of Nigeria's digitally connected population.

Appendix: Full Coarse Norm Theme Summary (Table A1)

The table below presents all 38 coarse norm themes identified through the pipeline, sorted by frequency. Scores represent averages across all beliefs within each theme.

Theme	N	R (W)	R (M)	B (W)	B (M)	R Diff	B Diff	%
Romantic Relationships	1812	3.11	2.65	2.32	1.99	+0.46	+0.33	5.8
Serious Relationships	1640	2.86	2.64	1.90	1.74	+0.21	+0.17	5.2
Sexual Ethics	1639	3.22	2.90	2.57	2.28	+0.31	+0.30	5.2
Family / Domestic	1475	3.15	2.70	2.32	2.01	+0.45	+0.31	4.7
Marriage as Goal	1426	3.09	2.68	2.32	1.96	+0.41	+0.36	4.5
Premarital Sex	1403	3.23	2.72	2.67	2.11	+0.51	+0.56	4.5
Islam and Gender	1318	3.78	2.93	3.30	2.52	+0.85	+0.79	4.2
Rape / Violence	1220	3.05	2.92	2.59	2.48	+0.13	+0.11	3.9
Marriage Roles	1150	2.95	2.74	2.12	1.94	+0.21	+0.18	3.7
Attractiveness	1070	3.14	2.18	2.13	1.49	+0.97	+0.64	3.4
Women in Family	1017	3.27	2.17	2.47	1.67	+1.10	+0.80	3.2
Financial Dynamics	1006	2.88	2.69	2.03	2.00	+0.19	+0.03	3.2
Inheritance / Paternity	919	3.05	2.56	2.06	1.77	+0.49	+0.29	2.9
Reproduction	914	3.34	2.53	2.54	1.90	+0.81	+0.64	2.9
Relationship Foundations	904	2.77	2.72	1.81	1.78	+0.06	+0.03	2.9
Marriage	895	3.07	2.67	2.34	1.98	+0.40	+0.35	2.8
Sexual Health	866	3.06	2.72	2.19	1.98	+0.35	+0.21	2.8
Female Gender Roles	830	2.98	2.13	2.32	1.68	+0.86	+0.65	2.6
Polygamy	813	3.23	2.49	2.59	1.89	+0.75	+0.70	2.6
Marriage Goal (2)	730	3.19	2.48	2.46	1.76	+0.71	+0.71	2.3
Domestic Violence	685	3.02	2.90	2.46	2.37	+0.12	+0.08	2.2
Partner Attributes	648	2.81	2.74	1.85	1.83	+0.07	+0.02	2.1
Promiscuity	614	2.99	2.81	2.33	2.16	+0.18	+0.17	2.0
Gender Stereotypes	602	3.12	2.49	2.56	2.02	+0.63	+0.54	1.9
Marital Gender Roles	601	3.24	2.41	2.60	1.93	+0.83	+0.67	1.9
Weddings	549	2.91	2.54	1.95	1.75	+0.37	+0.20	1.7
Parenting	547	2.80	2.74	1.85	1.82	+0.06	+0.03	1.7
Hijab	493	3.87	1.56	3.45	1.31	+2.31	+2.13	1.6
Gender Roles	487	2.99	2.67	2.07	1.94	+0.32	+0.13	1.5
Religion in Marriage	477	3.00	2.77	2.31	2.10	+0.23	+0.21	1.5
Gender & Employment	456	2.99	2.50	2.10	1.80	+0.49	+0.30	1.4
Modesty in Dress	440	3.26	2.40	2.54	1.79	+0.86	+0.76	1.4
Infidelity	423	3.06	2.77	2.51	2.23	+0.29	+0.28	1.3

Relationship Age	320	2.97	2.67	2.14	1.91	+0.30	+0.23	1.0
Homosexuality	312	3.62	3.60	3.37	3.36	+0.02	+0.01	1.0
Partner Criteria	287	2.78	2.77	1.83	1.83	+0.01	+0.00	0.9
Parental Responsibility	275	2.88	2.69	1.91	1.82	+0.19	+0.09	0.9
Relationships	193	2.84	2.73	1.92	1.85	+0.11	+0.07	0.6

R = Restrictiveness (1–5); B = Burden (1–5); Diff = Women – Men; % = share of total beliefs. N = number of beliefs in theme.

References and Resources

The methodology was informed by several social norms measurement frameworks, including the ALiGN platform for cross-country norms assessments, CARE Nigeria's norms diagnostic work, the UNDP Gender Social Norms Index, the G-Norm scale (EMERGE), the Breakthrough Action social norms taxonomy, and the World Values Survey gender module.